Optimization under uncertainty: stochastic programming

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Presentation Outline

1. Optimization under uncertainty
   - Why uncertainty changes the problem
   - Three approaches to uncertainty
   - Scoring an uncertain cost
   - Enforcing an uncertain constraint
   - Evaluating a decision
2. Stochastic Programming Approach
   - One-stage Problems
   - Two-stage Problems
   - The recourse value function
3. Information and discretization
   - Choosing the information structure
   - The value of information
   - Sample Average Approximation

Colour coding

Best effort:

- in the text, violet for definitions and red for key results and assumptions;
- in the maths, y_1, $\boldsymbol{y}_2$ decisions, ξ, $\boldsymbol{\xi}$ the uncertainty, λ dual variables, and y_1 a quantity held fixed.

After this session, you should be able to

- show that optimizing scenario by scenario, or at the average, does not necessarily produce a good decision;
- write the robust, stochastic and distributionally robust versions of a problem with uncertain data;
- score a random cost, or a constraint, by its expectation, value at risk, average value at risk or worst case;
- write a problem as a two-stage problem from its timing, its extensive formulation under finite support, and its recourse value function; say whether it has complete or relatively complete recourse;
- order the values of the nonadaptive, two-stage and perfect-information problems, and compute EVPI and VSS;
- evaluate a candidate decision out of sample, and say which way the value of a sampled problem is biased.

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Why is there uncertainty?

An optimization model needs data (costs, capacities, yields...), some of which is uncertain when the decision is taken. It may be

- measured imperfectly: a reaction yield, a pipe's resistance, a demand elasticity;
- in the future: tomorrow's wind, next week's demand, next year's electricity price;
- missing or corrupted, and the decision cannot wait;
- ...

A standard optimization problem

$$\begin{aligned} \min_{y_1 \in \mathbb{R}^n} \quad & \ell(y_1) \\ \text{s.t.} \quad & g(y_1) \leq 0 \end{aligned}$$

where

- y_1 is the decision (the subscript: first decision of the model);
- ℓ is the cost, or objective;
- $g(y_1) \leq 0$ are the constraints.

An optimization problem with uncertainty

Adding an uncertainty ξ:

$$\begin{aligned} \min_{y_1} \quad & \ell(y_1, \xi) \\ \text{s.t.} \quad & g(y_1, \xi) \leq 0 \end{aligned}$$

An optimization problem with uncertainty

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- ξ is unknown, and exogenous: the decision does not influence it.
- The cost is not well defined: for a fixed y_1, $\ell(y_1, \xi)$ is a function of ξ, not a number.
- The constraints are not well defined: $g(y_1, \xi) \leq 0$ has an unknown element ξ.

Optimizing at the average

Keep a single scenario, the mean: the expected value problem

$$v^{\mathrm{EV}} := \min_{y_1} \ \ell(y_1, \mathbb{E}_{\mathbb{P}}[\boldsymbol{\xi}]) \ ,$$

whose solution y_1^{EV} has expected cost $v^{\mathrm{EEV}} := \mathbb{E}_{\mathbb{P}}[\ell(y_1^{\mathrm{EV}}, \boldsymbol{\xi})]$, not v^{EV}.

A cyclist rides home on a road with traffic both ways; the pavement is safe but slow. He has had a drink: wherever he aims, he drifts left or right with equal probability. What does the expected value problem recommend?

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A cyclist rides home on a road with traffic both ways; the pavement is safe but slow. He has had a drink: wherever he aims, he drifts left or right with equal probability. What does the expected value problem recommend?
Aim at the painted line in the middle, where nothing drives. He is never on it: v^{EV} is achieved with probability 0, and the pavement is the answer.

Can't we just look at different values of the parameters?

- Natural idea: fix the uncertain parameters at a few values, the scenarios, and optimize for each: scenario-by-scenario optimization.
- Not enough, and can be misleading.
- Example:
  - an investor may buy, or sell short, 1000€ of a stock for a flat fee of 50€, or stand aside;
  - the stock gains or loses 10%, with equal probability;
  - ➥ what does optimizing each scenario recommend?

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Answer. Up: buy; down: sell short. Both scenarios promise a cost of -50, that is, a gain of 50, with opposite trades. In expectation either trade gains 0 and costs the fee, 50: stand aside, which no scenario recommends.

Common solution might not be relevant

A plant makes products A, B and C:

- shared capacity of 100 units, production cost 10 per unit;
- selling prices 11, 12 and 13;
- demand for A unlimited; B and C are substitutes, of total demand 100: ξ for B, $100 - \xi$ for C;
- production is planned before ξ is known; unsold units are lost.

1. For a given ξ, write the optimization problem;
2. What is, for each ξ, the optimal production plan?
3. Is there a plan that is optimal for every ξ?
4. If ξ is 0 or 100 with equal probability, what is the optimal plan? Comments?

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Answer. For a given ξ: C up to $100 - \xi$, B up to ξ, no A, at a cost $\xi - 300$; no plan is optimal for two values of ξ. The expected cost is $-y_A + 4\,y_B + 3.5\,y_C$: make only A, at -100, which appears in no scenario optimum. The average of the scenario plans, 50 of B and 50 of C, costs 375 on average: it is also the expected value solution, $v^{\mathrm{EV}} = -250$ and $v^{\mathrm{EEV}} = 375$.

Flexibility is not valued

- A buyer needs one unit of a good. Buying today costs 100.
- Waiting means paying an unknown price $p \in [50, 200]$.
- For 10, a trader offers the option (not the obligation) to buy later at 100.

1. What is the best decision when p is known in advance?
2. What is it when p is 50 or 200 with equal probability?

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1. What is the best decision when p is known in advance?
2. What is it when p is 50 or 200 with equal probability?

Answer.

1. Buy today if $p > 100$, wait otherwise: the option never helps.
2. Buying today costs 100, waiting 125 on average, and the option $10 + \frac{1}{2}(50) + \frac{1}{2}(100) = 85$: buy the option.

A scenario fixes the future, so it never values the right to change your mind.

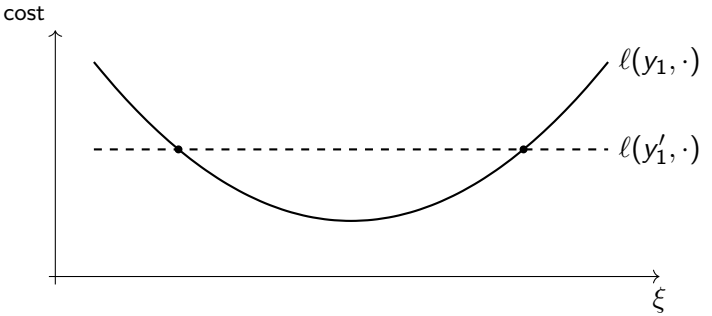
Neither the average scenario, nor the scenario solutions, nor their average recovers the optimal decision, and none values flexibility.

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Comparing two decisions

The cost $\ell(y_1, \xi)$ is a function of ξ, not a number: comparing two decisions means comparing two functions, which have no natural total order.



A model must collapse $\ell(y_1, \cdot)$ into one number, and say what $g(y_1, \xi) \leq 0$ requires.

Three approaches: a set, a law, a set of laws

- Robust Optimization (RO): $\xi \in \mathcal{R}$, a known uncertainty set, taken adversarially,

$$\min_{y_1} \sup_{\xi \in \mathcal{R}} \ell(y_1, \xi) \quad \text{s.t.} \quad g(y_1, \xi) \leq 0, \quad \forall \xi \in \mathcal{R} \, .$$

- Stochastic Programming (SP): $\xi \sim \mathbb{P}$, a random variable of known law,

$$\min_{y_1} \mathbb{E}_{\mathbb{P}}\big[\ell(y_1, \xi)\big] \quad \text{s.t.} \quad g(y_1, \xi) \leq 0, \quad \mathbb{P}\text{-a.s.}$$

- Distributionally Robust Optimization (DRO): the law is only assumed to lie in an ambiguity set $\mathcal{Q}$ of probability laws,

$$\min_{y_1} \sup_{\mathbb{Q} \in \mathcal{Q}} \mathbb{E}_{\mathbb{Q}}\big[\ell(y_1, \xi)\big] \, ,$$

the constraints holding $\mathbb{Q}$-a.s. for every $\mathbb{Q} \in \mathcal{Q}$.

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DRO extends both: $\mathcal{Q} = \{\mathbb{P}\}$ gives SP, and $\mathcal{Q} = \{\delta_\xi \mid \xi \in \mathcal{R}\}$, the Dirac masses on $\mathcal{R}$, gives RO.

Requirements and limits

- Stochastic programming:
  - needs a law $\mathbb{P}$ of $\boldsymbol{\xi}$, rarely known, estimated from data;
  - can be hard to solve: the expectation, an integral, is replaced by finitely many values, which blows up the dimension;
  - has dedicated methods (e.g. Benders decomposition).
- Robust optimization:
  - needs an uncertainty set $\mathcal{R}$;
  - can be overly conservative: the size of $\mathcal{R}$ largely decides how much;
  - has a complexity that depends strongly on $\mathcal{R}$.
- Distributionally robust optimization:
  - mixes the two: a stochastic program whose law is chosen robustly;
  - is a fast-growing field;
  - is still harder to implement;
  - typically takes $\mathcal{Q}$ a ball around the empirical law $\widehat{\mathbb{P}}_N$ of a sample, large enough to contain the true law $\mathbb{P}$ with high confidence.

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Requirements and limits

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The (deterministic) newsvendor problem

Each morning a newsboy buys papers at unit cost c and sells them during the day at unit price p, with $0 < c < p$; unsold papers are worthless. The day's demand is d papers.

How shall we model this?

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How shall we model this?

- Decision $y_1 \geq 0$, the papers bought, of which $\min(y_1, d)$ are sold
- Cost $\ell(y_1) = c\, y_1 - p \min(y_1, d)$

$$\min_{y_1 \geq 0} \quad c\, y_1 - p \min(y_1, d)\ ,$$

solved by $y_1^\star = d$: the one number nobody has in the morning.

The (stochastic) newsvendor problem

The demand is unknown when buying: a random variable $\boldsymbol{d}$ of known law.

- The stock $y_1 \geq 0$ stays a number, fixed in the morning for every outcome;
- the cost becomes random; we score it by its expectation.

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$$\min_{y_1 \geq 0} \quad \mathbb{E}_{\mathbb{P}}\big[c\, y_1 - p \min(y_1, \boldsymbol{d})\big]$$

Why the expectation?

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Why the expectation?
The law of large numbers: the newsvendor sells every day, and his average daily cost approaches the expected cost.

Solving the stochastic newsvendor problem

Proposition (Critical fractile)

Let $0 < c < p$ and $\boldsymbol{d} \geq 0$ have cdf F, and $F^{-1}(\alpha) := \inf\{t \mid F(t) \geq \alpha\}$. A stock $y_1^\star \geq 0$ is optimal if and only if

$$\mathbb{P}(\boldsymbol{d} < y_1^\star) \;\leq\; \frac{p-c}{p} \;\leq\; \mathbb{P}(\boldsymbol{d} \leq y_1^\star)\,,$$

and $F^{-1}\big((p-c)/p\big)$ is one. If F is continuous: $F(y_1^\star) = (p-c)/p$.

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Proof. $J(y_1) := \mathbb{E}_{\mathbb{P}}\big[c\,y_1 - p\min(y_1, \boldsymbol{d})\big]$ is finite, since $\min(y_1, \boldsymbol{d}) \in [0, y_1]$, and convex; by dominated convergence

$$J'_+(y_1) = c - p\,\mathbb{P}\big(\boldsymbol{d} > y_1\big), \qquad J'_-(y_1) = c - p\,\mathbb{P}\big(\boldsymbol{d} \geq y_1\big).$$

Optimality is $J'_-(y_1^\star) \leq 0 \leq J'_+(y_1^\star)$, and

$$J'_-(y_1^\star) \leq 0 \iff \mathbb{P}\big(\boldsymbol{d} < y_1^\star\big) \leq \tfrac{p-c}{p}, \qquad J'_+(y_1^\star) \geq 0 \iff \mathbb{P}\big(\boldsymbol{d} \leq y_1^\star\big) \geq \tfrac{p-c}{p}$$

(at 0, only $J'_+ \geq 0$ is needed, and $\mathbb{P}(\boldsymbol{d} < 0) = 0$). $F^{-1}\big((p-c)/p\big)$ satisfies it by right continuity of F. □

The robust newsvendor problem

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$$\min_{y_1 \geq 0} \quad \max_{d \in [\underline{d}, \overline{d}]} c\, y_1 - p \min(y_1, d)$$

The inner cost is nonincreasing in d, so the maximum is at $d = \underline{d}$ for every y_1:

$$\min_{y_1 \geq 0} \quad c\, y_1 - p \min(y_1, \underline{d}) \ ,$$

the deterministic problem at the lower bound: $y_1^\star = \underline{d}$.

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Alternative cost functions I

- For a random cost $\ell(y_1, \xi)$, it is natural to minimize $\mathbb{E}_{\mathbb{P}}[\ell(y_1, \xi)]$.
- Justified by the law of large numbers when the decision is repeated many times and only the total matters: the average cost paid approaches the expectation.
  - ➥ The newsvendor sells 250 days a year: is the expectation his score? And if he pays rent every week?

- But the expectation may not reflect your attitude to risk:
  - Pay 1€ for one chance in ten of winning 10€? And 1000€ for one chance in ten of winning 10 000€?
  - To catch a flight within the hour: one route takes 50 minutes for sure; another 40 minutes with probability 4/5, 70 minutes with probability 1/5. Which one?

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  - ➥ The newsvendor sells 250 days a year: is the expectation his score? And if he pays rent every week? Living at home, yes: the year's cost is close to 250 times the expected daily cost. Owing rent on Friday, no: a bad week is not made up by later good ones, so score the tail.
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Alternative cost functions II

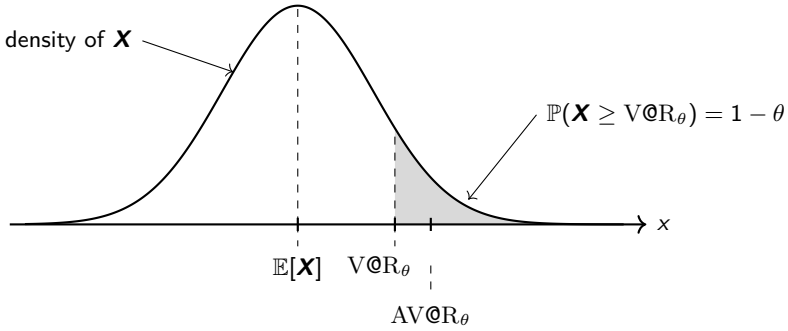
A risk measure ρ collapses the random cost into one number: $\min_{y_1} \rho(\ell(y_1, \boldsymbol{\xi}))$. Besides $\rho = \mathbb{E}_{\mathbb{P}}$:

- threshold probability, $\mathbb{P}(\ell(y_1, \boldsymbol{\xi}) > 0)$: how often, not by how much;
- Value-at-Risk at level $\theta \in [0, 1]$ of a random cost $\boldsymbol{X}$, $\mathrm{V@R}_\theta(\boldsymbol{X}) := \inf\{t \mid \mathbb{P}(\boldsymbol{X} \leq t) \geq \theta\}$, the smallest cost exceeded with probability at most $1 - \theta$;
- Average Value-at-Risk, the mean cost over the worst fraction $1 - \theta$ of outcomes; for $\theta < 1$ and $\boldsymbol{X}$ with a density and a finite expectation, $\mathrm{AV@R}_\theta(\boldsymbol{X}) := \mathbb{E}_{\mathbb{P}}[\boldsymbol{X} \mid \boldsymbol{X} > \mathrm{V@R}_\theta(\boldsymbol{X})]$.[1] From the expectation ($\theta = 0$) to the worst case ($\theta \to 1$).[2]
- Not recommended: the expectation plus a multiple of the variance (Markowitz), which is not monotone: a cost that is never positive can score above a cost that is always zero.

[1] For $0 < \theta < 1$ and a cost of finite expectation, of any law, scenarios included: $\mathrm{AV@R}_\theta(\boldsymbol{X}) = \min_{\eta \in \mathbb{R}} \left\{\eta + \frac{1}{1-\theta} \mathbb{E}_{\mathbb{P}}\left[\left(\boldsymbol{X} - \eta\right)^+\right]\right\}$, attained at $\eta = \mathrm{V@R}_\theta(\boldsymbol{X})$ (Rockafellar and Uryasev).

[2] Some authors use the complementary level: $\mathrm{AV@R}_\alpha$ averages the worst fraction α of the outcomes, $\alpha = 1 - \theta$.

Value at risk and average value at risk



A Gaussian cost, $\theta = 0.9$. The value at risk is exceeded on the shaded tail, of probability $1 - \theta$; the average value at risk is the tail's mean, hence to its right.

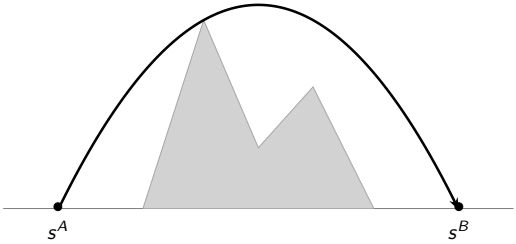
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A drone in a corridor

A drone flies from s^A at date 0 to s^B at date T; its position $s_t \in \mathbb{R}^2$ must stay in the authorized airspace $\mathcal{X}$, closed and not the whole plane. The decision is the displacement y_t over period t, of norm at most Δ, and energy grows with its square:

$$
\begin{aligned}
\min_{(y_t)_{t\in[T]}} \quad & \sum_{t=1}^{T} \|y_t\|^2 \\
\text{s.t.} \quad & s_t = s_{t-1} + y_t, && \forall t \in [T] \\
& s_t \in \mathcal{X}, && \forall t \in [T] \\
& \|y_t\| \leq \Delta, && \forall t \in [T] \\
& s_0 = s^A, \quad s_T = s^B\ .
\end{aligned}
$$



Alternative constraints

- Extending $g(y_1, \xi) \leq 0$ to $g(y_1, \boldsymbol{\xi}) \leq 0$ $\mathbb{P}$-a.s. can be extremely conservative, often infeasible.
- Example. Imprecise propulsion: over period t the drone covers $y_t + \xi_t$, with $\xi_t \sim \mathcal{N}(0, \sigma^2 I_2)$ independent over t, and the commands are fixed in advance. Required almost surely, the problem is infeasible, twice over:
  - s_T has a law with a density, so $\mathbb{P}(s_T = s^B) = 0$;
  - even with a landing zone: a Gaussian charges the whole plane and the complement of $\mathcal{X}$ is open and nonempty, so every plan leaves $\mathcal{X}$ with positive probability, at every period.
- The deviation from the plan at period t has standard deviation $\sigma\sqrt{t}$: a corridor of $10\sigma\sqrt{T}$ on either side of the planned path is left with probability below 10^{-21} at every period. Incredibly small, but not zero.

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Alternative constraints II

Five constraints from one requirement, each of the form $\rho(g(y_1, \xi)) \leq 0$:

- $\mathbb{E}_{\mathbb{P}}[g(y_1, \xi)] \leq 0$, in expectation: quality-of-service constraints, where a violation in one outcome is offset by slack in another;
- $\mathbb{P}(g(y_1, \xi) \leq 0) \geq \theta$, a chance constraint, conservative for θ close to 1: easy to state, but silent on what happens when it is violated;
- $\mathrm{AV@R}_\theta(g(y_1, \xi)) \leq 0$, by a risk measure;
- $g(y_1, \xi) \leq 0$, $\mathbb{P}$-a.s., almost surely: right when the support is bounded and the requirement physical;
- $g(y_1, \xi) \leq 0$, $\forall \xi \in \mathcal{R}$, robustly: the almost-sure constraint with the support replaced by a set you choose (for $g(y_1, \cdot)$ continuous, the two coincide when $\mathcal{R} = \mathrm{supp}(\xi)$).

- The chance constraint is $\mathrm{V@R}_\theta(g(y_1, \xi)) \leq 0$, whatever the law.
- The $\mathrm{AV@R}_\theta$ constraint implies it, since $\mathrm{AV@R}_\theta \geq \mathrm{V@R}_\theta$, and is convex whenever $g(\cdot, \xi)$ is.

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Computing expectation

- Computing $\mathbb{E}_{\mathbb{P}}\left[\ell(y_1, \boldsymbol{\xi})\right]$, even for a given y_1, is costly:
  - with a density, a multidimensional integral;
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Solution: sampling, justified by the law of large numbers (LLN) and the central limit theorem (CLT).

- Draw N independent realizations $(\xi^n)_{n\in[N]}$ of $\boldsymbol{\xi}$;
- compute the sample average $\dfrac{1}{N}\sum_{n=1}^{N} \ell(y_1, \xi^n)$;
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This is the Monte Carlo method (details with the sample average approximation).

Consequence: evaluating a decision is difficult

- Even evaluating a solution can require approximation.
- So can checking admissibility: a chance constraint is the expectation of an indicator, an almost-sure constraint a statement about every outcome.
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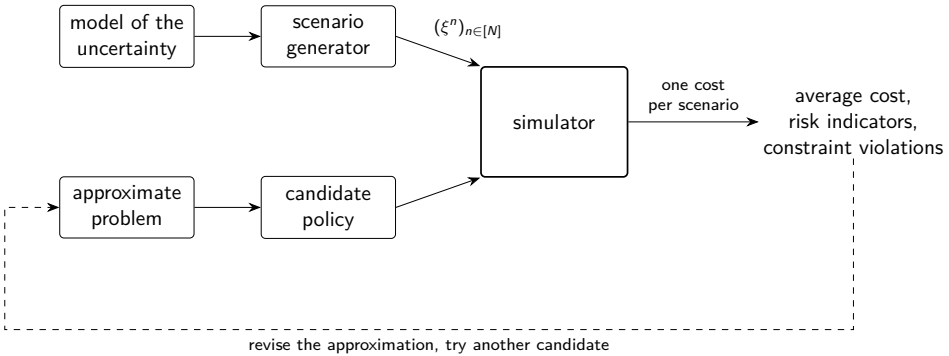
Optimization problem and simulator

- Stochastic optimization problems are often not well posed, and must be approximated before being solved.
- Good practice: write a simulator, code that takes realizations of the uncertainty and a policy (a rule turning the available information into a decision), and returns the costs and whatever else is worth watching.

Three steps:

1. define the simulator first, before solving any model: it defines what a good decision is;
2. obtain a candidate policy by solving an approximate problem;
3. evaluate it on the simulator, on more than the average cost (spread, constraint violations...), and if it is not good enough, go back to step 2 with a different approximation.

Evaluating a candidate on the simulator



The scenario generator produces realizations, the approximate problem a policy; the simulator runs one against the other and returns one cost per scenario, from which indicators are estimated. The dashed arrow goes back to the approximate problem, not to the uncertainty model.

Judge a candidate out of sample

- The sample that produced a candidate cannot judge it: a decision minimizing the average cost over $(\xi^n)_{n\in[N]}$ is fitted to those draws, and its average on them is optimistic on average.
- Out-of-sample: evaluate on a fresh sample, never shown to the optimizer; an unbiased estimate of its expected cost under the sampled law.
- Backtest: evaluate on held-out data, such as historical records; this tests the law itself, misspecified or shifted since the fit.
- Compare two methods by running both policies on one common out-of-sample set, never by the optimal values they report.

Example. Newsvendor, $c = 1$, $p = 2$, demand uniform on $[0, 100]$: the optimal stock is 50, of expected cost -25. Fitted to three days of demand 30, 60, 80, the stock is 60: it costs -40 on those days, and -24 in expectation.

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Conclusion

Facing an optimization problem under uncertainty, decide

- how to model the uncertainty (a set, a random variable, a set of laws);
- how to score a random cost, i.e. your risk preference (expectation, value at risk. . .);
- what a constraint requires when it cannot hold in every outcome;
- what information is available for each decision: the information structure (later);
- and set up a simulator to evaluate your solutions.

Course plan

1. Stochastic programming formulation
2. Robust optimization: introduction, approaches and examples
3. Decomposition methods for two-stage stochastic programming
4. Methods for multistage stochastic programming
5. Advanced methods for robust optimization
6. Exam

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Finitely supported noise

Assumption (FSN, finitely supported noise)

$\boldsymbol{\xi}$ has a finite support, enumerated without repetition by $\Omega := [S] = \{1, \dots, S\}$,

$$\mathbb{P}(\boldsymbol{\xi} = \xi^\omega) = \pi^\omega > 0, \qquad \omega \in \Omega\,.$$

We call ξ^ω a scenario and π^ω its probability.[a]

[a] A continuous law can be sampled, and the sample used instead: the Sample Average Approximation, at the end of this lecture.

One-Stage Problems

Proposition

Under (FSN), the one-stage stochastic program

$$\begin{aligned} \min_{y_1 \in \mathbb{R}^{n_1}} \quad & \mathbb{E}_{\mathbb{P}}\Big[\ell(y_1, \boldsymbol{\xi})\Big] \\ s.t. \quad & g(y_1, \boldsymbol{\xi}) \leq 0, \qquad \mathbb{P}\text{-}a.s. \end{aligned}$$

is equivalent to the deterministic problem

$$\begin{aligned} \min_{y_1 \in \mathbb{R}^{n_1}} \quad & \sum_{\omega \in \Omega} \pi^{\omega} \ell(y_1, \xi^{\omega}) \\ s.t. \quad & g(y_1, \xi^{\omega}) \leq 0, \qquad \forall \omega \in \Omega \, . \end{aligned}$$

Newsvendor problem (continued)

The demand takes the values $(d^\omega)_{\omega\in\Omega}$ with probabilities $(\pi^\omega)_{\omega\in\Omega}$.

Newsvendor problem (continued)

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The stochastic newsvendor problem reads

$$\min_{y_1\geq 0} \quad \sum_{\omega\in\Omega} \pi^\omega \Big(c\, y_1 - p \min(y_1, d^\omega) \Big) \,,$$

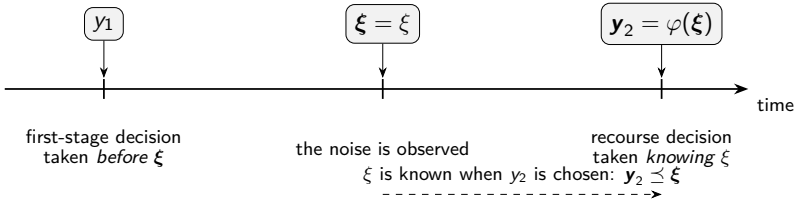
convex and piecewise affine in one variable, with a breakpoint at each demand scenario.

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Recourse

Usually we can act again once the uncertainty is revealed: a second decision y_2 corrects the first, $y_1 \rightsquigarrow \boldsymbol{\xi} \rightsquigarrow y_2$.

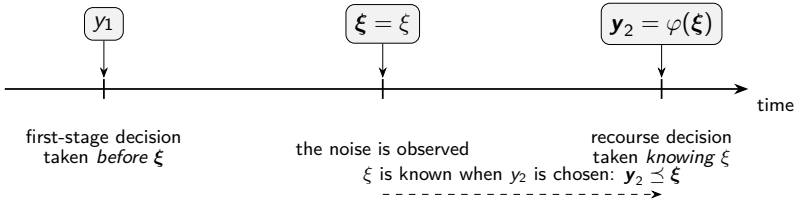


If it may depend on the observation, then $\boldsymbol{y}_2 = \varphi(\boldsymbol{\xi})$ for some φ on $\Xi := \mathrm{supp}(\boldsymbol{\xi})$: the recourse is a random variable, measurable with respect to $\boldsymbol{\xi}$, written

$$\boldsymbol{y}_2 \preceq \boldsymbol{\xi} \ .$$

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Two-stage Problem

The recourse $\boldsymbol{y}_2$ being a random variable, the two-stage stochastic program reads, in compact form,

$$\begin{array}{ll} \min\limits_{y_1, \boldsymbol{y}_2} & \mathbb{E}_{\mathbb{P}}\Big[\ell(y_1, \boldsymbol{\xi}, \boldsymbol{y}_2)\Big] \\ \text{s.t.} & g(y_1, \boldsymbol{\xi}, \boldsymbol{y}_2) \leq 0, \qquad \mathbb{P}\text{-a.s.} \\ & \boldsymbol{y}_2 \preceq \boldsymbol{\xi} \end{array}$$

over $y_1 \in \mathbb{R}^{n_1}$ deterministic and $\boldsymbol{y}_2$ random, with values in $\mathbb{R}^{n_2}$.

- y_1 is the first-stage decision;
- $\boldsymbol{y}_2$ is the second-stage (or recourse) decision.

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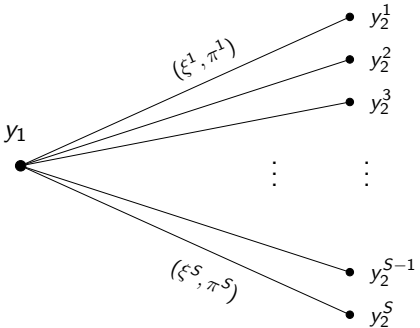
Extensive formulation

Proposition (Extensive formulation)

Under (FSN), the two-stage program is equivalent to the deterministic problem

$$\min_{y_1,(y_2^\omega)_{\omega\in\Omega}} \quad \sum_{\omega\in\Omega} \pi^\omega \ell(y_1,\xi^\omega,y_2^\omega)$$
$$s.t. \quad g(y_1,\xi^\omega,y_2^\omega)\le 0, \qquad \forall \omega\in\Omega\ ,$$

in $n_1 + S\,n_2$ variables, with $y_2^\omega = \boldsymbol{y}_2$ on the event $\{\boldsymbol{\xi} = \xi^\omega\}$.



A deterministic problem, the extensive formulation: give it to a solver first; decomposition methods when it is too large.

Two-stage newsvendor problem I

The newsvendor as a two-stage problem:

- y_1, the papers bought in the morning
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- y_1, the papers bought in the morning $\rightsquigarrow$ first-stage decision;
- y_2, the papers sold during the day $\rightsquigarrow$ second-stage decision.

$$\begin{aligned} \min_{y_1 \geq 0,\, \boldsymbol{y_2} \geq 0} \quad & \mathbb{E}_{\mathbb{P}}\left[c\, y_1 - p\, \boldsymbol{y_2}\right] \\ \text{s.t.} \quad & \boldsymbol{y_2} \leq y_1, \quad \boldsymbol{y_2} \leq \boldsymbol{d}, \qquad \mathbb{P}\text{-a.s.} \\ & \boldsymbol{y_2} \preceq \boldsymbol{d} \end{aligned}$$

Sales are bounded by the stock and the demand, and depend on the demand; the purchase does not.

Two-stage newsvendor problem II

In extensive formulation:

$$\begin{aligned} \min_{y_1,(y_2^\omega)_{\omega\in\Omega}} \quad & \sum_{\omega\in\Omega} \pi^\omega \left(c\, y_1 - p\, y_2^\omega\right) & \\ \text{s.t.} \quad & 0 \leq y_2^\omega \leq y_1 & \forall \omega \in \Omega \\ & y_2^\omega \leq d^\omega & \forall \omega \in \Omega \end{aligned}$$

A linear program in $1 + S$ variables: one second-stage decision y_2^ω per demand scenario, a single first-stage decision y_1.

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Stage decomposition

Suppose the cost and constraints split as

$$\ell(y_1, \xi, y_2) = \ell_1(y_1) + \ell_2(y_1, \xi, y_2),$$
$$g(y_1, \xi, y_2) \leq 0 \iff \Big(g_1(y_1) \leq 0 \quad \text{and} \quad g_2(y_1, \xi, y_2) \leq 0\Big).$$

A modelling choice: a split always exists ($\ell_1 \equiv 0$, $g_1 \equiv 0$), but is not unique.

The recourse value function is the value of the second-stage problem

$$\dot{V}_2(y_1, \xi) := \inf_{y_2 \in \mathbb{R}^{n_2}} \quad \ell_2(y_1, \xi, y_2)$$
$$\text{s.t.} \quad g_2(y_1, \xi, y_2) \leq 0 \ ,$$

with $\inf \emptyset = +\infty$; its expectation is $V_2(y_1) := \mathbb{E}_{\mathbb{P}}\big[\dot{V}_2(y_1, \xi)\big]$ (the dot: still a function of the noise).

(BB) Bounded below. For every y_1 with $g_1(y_1) \leq 0$ and every $\xi \in \Xi$, $\dot{V}_2(y_1, \xi) > -\infty$.

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Stage decomposition theorem

Theorem (Stage decomposition)

Under (FSN), (BB) and a split, the two-stage program is equivalent to the first-stage problem

$$\begin{aligned} \min_{y_1 \in \mathbb{R}^{n_1}} \quad & \ell_1(y_1) + V_2(y_1) \\ s.t. \quad & g_1(y_1) \leq 0 \,, \end{aligned}$$

in the sense that

- *the two problems have the same value;*
- *the first stage of a two-stage optimum is optimal for the first-stage problem;*
- *conversely, an optimum $y_1^\star$ of the first-stage problem is the first stage of a two-stage optimum, provided the infima defining $\dot{V}_2(y_1^\star, \xi^\omega)$ are attained.*

Proof idea. In the extensive formulation, minimize first over $(y_2^\omega)_{\omega \in \Omega}$ at fixed y_1: the problem separates, with value $\sum_{\omega \in \Omega} \pi^\omega \dot{V}_2(y_1, \xi^\omega) = V_2(y_1)$.

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Admissible sets

For a given split, we set

$$\mathcal{Y}_1 := \{y_1 \in \mathbb{R}^{n_1} \mid g_1(y_1) \leq 0\}, \qquad \mathcal{Y}_2(y_1, \xi) := \{y_2 \in \mathbb{R}^{n_2} \mid g_2(y_1, \xi, y_2) \leq 0\} .$$

- $\mathcal{Y}_2(y_1, \xi)$ is the admissible set of the second-stage problem; $\dot{V}_2(y_1, \xi) = +\infty$ exactly when it is empty;
- $\mathcal{Y}_1$ only contains the admissible set of the first-stage problem.

Proposition

Under (FSN) and (BB), the first-stage decisions of finite cost are

$$\mathcal{Y}_1^{\text{feas}} := \{y_1 \in \mathcal{Y}_1 \mid \ell_1(y_1) + V_2(y_1) < +\infty\} = \{y_1 \in \mathcal{Y}_1 \mid \mathbb{P}(\mathcal{Y}_2(y_1, \xi) \neq \emptyset) = 1\} \subseteq \mathcal{Y}_1 ,$$

and the inclusion may be strict.

Admissible sets

For a given split, we set

$$\mathcal{Y}_1 := \{y_1 \in \mathbb{R}^{n_1} \mid g_1(y_1) \leq 0\}, \qquad \mathcal{Y}_2(y_1, \xi) := \{y_2 \in \mathbb{R}^{n_2} \mid g_2(y_1, \xi, y_2) \leq 0\} .$$

- $\mathcal{Y}_2(y_1, \xi)$ is the admissible set of the second-stage problem; $\dot{V}_2(y_1, \xi) = +\infty$ exactly when it is empty;
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Complete and relatively complete recourse

- Complete recourse: every $y_1 \in \mathbb{R}^{n_1}$ has, almost surely, an admissible recourse:

$$\mathbb{P}(\mathcal{Y}_2(y_1, \xi) \neq \emptyset) = 1, \quad \forall y_1 \in \mathbb{R}^{n_1}.$$

- Relatively complete recourse: every $y_1 \in \mathcal{Y}_1$ has, almost surely, an admissible recourse:

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Obtaining relatively complete recourse

Take a two-stage program in first-stage form

$$\min_{y_1 \in \mathcal{Y}_1} \Big\{ \ell_1(y_1) + V_2(y_1) \Big\} \qquad \text{with} \quad \dot{V}_2(y_1, \xi) := \inf_{y_2 \in \mathcal{Y}_2(y_1, \xi)} \ell_2(y_1, \xi, y_2),$$

not necessarily with relatively complete recourse. The induced constraints are

$$\mathcal{Y}_1^{\text{ind}} := \Big\{ y_1 \in \mathbb{R}^{n_1} \quad | \quad \mathbb{P}\big(\mathcal{Y}_2(y_1, \boldsymbol{\xi}) \neq \emptyset\big) = 1 \Big\}.$$

Proposition (Induced-constraint reformulation)

Assume (FSN), (BB), and a finite value. Then

$$\inf_{y_1 \in \mathcal{Y}_1} \Big\{ \ell_1(y_1) + V_2(y_1) \Big\} = \inf_{y_1 \in \mathcal{Y}_1 \cap \mathcal{Y}_1^{\text{ind}}} \Big\{ \ell_1(y_1) + V_2(y_1) \Big\},$$

with the same optimal solutions, if any, and the right-hand problem has relatively complete recourse, with admissible set $\mathcal{Y}_1 \cap \mathcal{Y}_1^{\text{ind}}$.

Obtaining relatively complete recourse

Take a two-stage program in first-stage form

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Two-stage framework: three information structures

In $\min_{\boldsymbol{y}_1, \boldsymbol{y}_2} \mathbb{E}_{\mathbb{P}}[\ell(\boldsymbol{y}_1, \boldsymbol{\xi}, \boldsymbol{y}_2)]$, an information structure constrains the decisions, not the data (cost, constraints and law of $\boldsymbol{\xi}$ unchanged).

- Nonadaptive: y_1 and y_2 deterministic, both chosen without knowing $\boldsymbol{\xi}$. A small decision set; specific methods (e.g. stochastic gradient).
- Two-stage: y_1 deterministic, $\boldsymbol{y}_2 \preceq \boldsymbol{\xi}$. The problem of stochastic programming.
- Perfect information: $\boldsymbol{y}_1 \preceq \boldsymbol{\xi}$ and $\boldsymbol{y}_2 \preceq \boldsymbol{\xi}$. One deterministic problem per outcome, then the expectation of their values.[3]

Nonadaptive



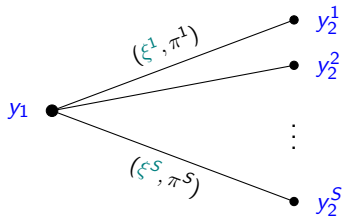
[3] Also called *open-loop* (nonadaptive) and *anticipative* (perfect information). First-stage and recourse decisions are often called *here-and-now* and *wait-and-see* (Shapiro et al.); the wait-and-see value WS of Birge and Louveaux is the perfect-information value.

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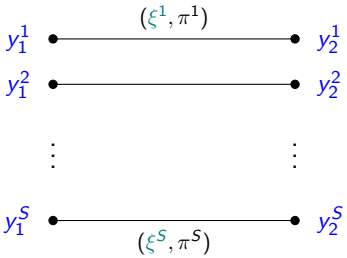
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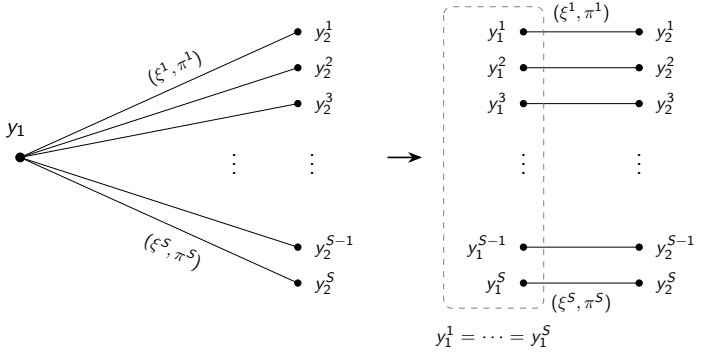
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Extensive and split formulations



The same program twice. Left, the extensive formulation: the S scenarios form one problem through the shared root decision. Right, the split formulation: S deterministic problems of the usual size, joined only by the boxed constraint.

Information structures for the newsvendor I

Nonadaptive:

$$\begin{aligned} \min_{y_1, y_2} \quad & \sum_{\omega \in \Omega} \pi^\omega (c\, y_1 - p\, y_2) \\ \text{s.t.} \quad & 0 \leq y_2 \leq y_1 \\ & y_2 \leq d^\omega \qquad \forall \omega \in \Omega \end{aligned}$$

Information structures for the newsvendor II

Two-stage:

$$\min_{y_1,(y_2^\omega)_{\omega\in\Omega}} \quad \sum_{\omega\in\Omega} \pi^\omega (c\, y_1 - p\, y_2^\omega)$$

$$\text{s.t.} \quad 0 \le y_2^\omega \le y_1 \qquad \forall \omega \in \Omega$$

$$y_2^\omega \le d^\omega \qquad \forall \omega \in \Omega$$

Information structures for the newsvendor III

Perfect information:

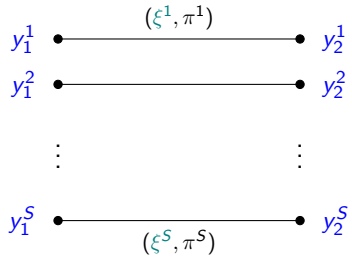
$$\begin{aligned} \min_{(y_1^\omega, y_2^\omega)_{\omega\in\Omega}} \quad & \sum_{\omega\in\Omega} \pi^\omega \left(c\, y_1^\omega - p\, y_2^\omega\right) \\ \text{s.t.} \quad & 0 \leq y_2^\omega \leq y_1^\omega && \forall \omega \in \Omega \\ & y_2^\omega \leq d^\omega && \forall \omega \in \Omega \end{aligned}$$

Comparing the information structures

The three structures in one formulation:

$$\min_{(y_1^\omega, y_2^\omega)_{\omega\in\Omega}} \quad \sum_{\omega\in\Omega} \pi^\omega \ell\left(y_1^\omega, \xi^\omega, y_2^\omega\right)$$
$$\text{s.t.} \quad g\left(y_1^\omega, \xi^\omega, y_2^\omega\right) \leq 0 \qquad \forall \omega \in \Omega$$

Perfect information



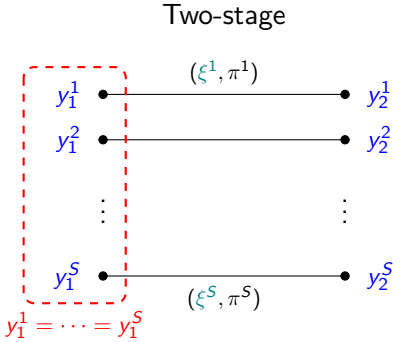
Each added constraint shrinks the feasible set, so more information relaxes: with v^{NA}, v^{SP}, v^{PI} the values of the nonadaptive, two-stage and perfect-information models,

$$v^{\mathrm{PI}}$$

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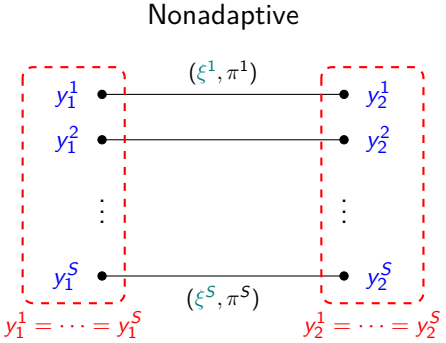
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Value of information

- Expected value of perfect information, $\mathrm{EVPI} := v^{\mathrm{SP}} - v^{\mathrm{PI}} \geq 0$: what knowing ξ in advance would save, hence a ceiling on what any forecast can be worth.
- Expected value problem: the noise replaced by its mean,

$$v^{\mathrm{EV}} := \min_{y_1, y_2} \{\ell(y_1, \mathbb{E}_{\mathbb{P}}[\xi], y_2) \mid g(y_1, \mathbb{E}_{\mathbb{P}}[\xi], y_2) \leq 0\},$$

  with y_1^{EV} the first stage of a solution.
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Comparing the values

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$$v^{\mathrm{PI}} \leq v^{\mathrm{SP}} \leq v^{\mathrm{NA}} \qquad \text{and} \qquad v^{\mathrm{SP}} \leq v^{\mathrm{EEV}},$$
- v^{EEV} and v^{NA} are not comparable, the recourse being free in v^{EEV}: the newsvendor below has $v^{\mathrm{EEV}} < v^{\mathrm{NA}}$; the production plan, with no recourse, has $v^{\mathrm{NA}} = v^{\mathrm{SP}} = -100 < v^{\mathrm{EEV}} = 375$.

The newsvendor with $c = 1$, $p = 3$ and a demand of 0 or 100, equiprobable:

v^{PI}	v^{EV}	v^{SP}	v^{EEV}	v^{NA}	EVPI	VSS
-100	-100	-50	-25	0	50	25

The stochastic stock is 100, the expected value stock 50.

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Convexity in the right-hand side

- A linear two-stage program with second stage

$$\dot{V}_2(y_1, \xi) = \min_{y_2 \geq 0} \left\{ c_2^\top y_2 \mid C_2\, y_2 = d_2(\xi) - B_2(\xi)\, y_1 \right\},$$

with fixed recourse (C_2 and c_2 deterministic), d_2 and B_2 affine in ξ, $\boldsymbol{\xi}$ integrable, and a nonempty dual set $D := \left\{ \lambda \mid C_2^\top \lambda \leq c_2 \right\}$. Then $\dot{V}_2(y_1, \xi) = \sup_{\lambda \in D} \lambda^\top \left(d_2(\xi) - B_2(\xi) y_1 \right)$ is convex in ξ; by Jensen,

$$V_2(y_1) \geq \dot{V}_2\left(y_1, \mathbb{E}_{\mathbb{P}}\left[\boldsymbol{\xi}\right]\right) \quad \text{for every } y_1 \,.$$

- Minimizing over y_1: under these hypotheses,

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Solving the problems

- y_1^{EV} is easy (one deterministic problem), but v^{EEV} needs a second-stage solve per scenario (or per drawn scenario, by Monte Carlo).
- v^{NA}: one deterministic problem under (FSN); otherwise specific methods (e.g. stochastic gradient).
- v^{SP}: stochastic-programming methods, of two main kinds:
  - Lagrangian decomposition (e.g. progressive hedging);
  - Benders decomposition (e.g. the L-shaped method, nested decomposition).
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- y_1^{EV} is easy (one deterministic problem), but v^{EEV} needs a second-stage solve per scenario (or per drawn scenario, by Monte Carlo).
- v^{NA}: one deterministic problem under (FSN); otherwise specific methods (e.g. stochastic gradient).
- v^{SP}: stochastic-programming methods, of two main kinds:
  - Lagrangian decomposition (e.g. progressive hedging);
  - Benders decomposition (e.g. the L-shaped method, nested decomposition).
- v^{PI}: one deterministic problem per scenario; with too many, Monte Carlo: solve for each drawn ξ^n and average the values.
- v^{EV}: easy to compute, but a bound only under convexity.

Presentation Outline

1. Optimization under uncertainty
   - Why uncertainty changes the problem
   - Three approaches to uncertainty
   - Scoring an uncertain cost
   - Enforcing an uncertain constraint
   - Evaluating a decision
2. Stochastic Programming Approach
   - One-stage Problems
   - Two-stage Problems
   - The recourse value function
3. Information and discretization
   - Choosing the information structure
   - The value of information
   - Sample Average Approximation

How to deal with continuous distributions?

Under finite support, the two-stage problem

$$\begin{aligned} \min_{y_1, \boldsymbol{y}_2} \quad & \mathbb{E}_{\mathbb{P}}\Big[\ell(y_1, \boldsymbol{\xi}, \boldsymbol{y}_2)\Big] \\ \text{s.t.} \quad & g(y_1, \boldsymbol{\xi}, \boldsymbol{y}_2) \leq 0, \qquad \mathbb{P}\text{-a.s.} \\ & \boldsymbol{y}_2 \preceq \boldsymbol{\xi} \end{aligned}$$

becomes

$$\begin{aligned} \min_{y_1, (y_2^\omega)_{\omega\in\Omega}} \quad & \sum_{\omega\in\Omega} \pi^\omega \ell(y_1, \xi^\omega, y_2^\omega) \\ \text{s.t.} \quad & g(y_1, \xi^\omega, y_2^\omega) \leq 0, \qquad \forall \omega \in \Omega. \end{aligned}$$

For a continuous law (e.g. Gaussian), the extensive formulation would need an infinite number of recourse variables, one per outcome.

Simplest idea: sample ξ

First-stage problem: $v^{\mathrm{SP}} = \inf_{y_1 \in \mathcal{Y}_1} J(y_1)$, with

$$J(y_1) := \ell_1(y_1) + \mathbb{E}_{\mathbb{P}}\left[\dot{V}_2(y_1, \xi)\right] .$$

- Draw independent copies $(\xi^n)_{n\in[N]}$ of ξ, of empirical distribution $\widehat{\mathbb{P}}_N := \frac{1}{N}\sum_{n=1}^N \delta_{\xi^n}$.
- Replacing $\mathbb{P}$ by $\widehat{\mathbb{P}}_N$ gives an (FSN) problem, with N scenarios of probability $1/N$, solved by its extensive formulation: the sample average approximation

$$\widehat{v}_N := \inf_{y_1 \in \mathcal{Y}_1} \widehat{J}_N(y_1) , \qquad \widehat{J}_N(y_1) := \ell_1(y_1) + \frac{1}{N}\sum_{n=1}^N \dot{V}_2(y_1, \xi^n) ,$$

  with $\widehat{y}_1^N$ an optimal solution.
- The sample has law $\mathbb{P}^{\otimes N}$, so $\widehat{\mathbb{P}}_N$, $\widehat{J}_N$, $\widehat{v}_N$ and $\widehat{y}_1^N$ are random; $\mathbb{E}_{\mathbb{P}^{\otimes N}}$ is the expectation over the draw.

Estimating the cost of a fixed decision

- For y_1 fixed *before* the draw, each ξ^n has the law of ξ, so

$$\mathbb{E}_{\mathbb{P}^{\otimes N}}\left[\widehat{J}_N(y_1)\right] = J(y_1) \qquad \text{for every } y_1 \in \mathcal{Y}_1 \text{ with } J(y_1) \in \mathbb{R}\,:$$

  $\widehat{J}_N(y_1)$ is an unbiased estimator of $J(y_1)$.

- Recall (CLT). If $(\boldsymbol{C}_n)_{n\in\mathbb{N}}$ are independent, identically distributed, with finite variance, then, in law,

$$\sqrt{N}\Big(\frac{1}{N}\sum_{n=1}^{N}\boldsymbol{C}_n - \mathbb{E}[\boldsymbol{C}_1]\Big) \Longrightarrow \mathcal{N}\big(0, \mathrm{Var}[\boldsymbol{C}_1]\big)\,.$$

Estimating the cost of a fixed decision: confidence interval

- With $\boldsymbol{C}_n := \dot{V}_2(y_1, \xi^n)$, their sample mean $\bar{V}_N := \frac{1}{N}\sum_{n=1}^{N} \boldsymbol{C}_n$ and empirical variance $\hat{\sigma}^2 := \frac{1}{N-1}\sum_{n=1}^{N} \left(\boldsymbol{C}_n - \bar{V}_N\right)^2$,

$$\mathbb{P}^{\otimes N}\left(J(y_1) \in \left[\widehat{J}_N(y_1) - \frac{z\,\hat{\sigma}}{\sqrt{N}},\ \widehat{J}_N(y_1) + \frac{z\,\hat{\sigma}}{\sqrt{N}}\right]\right) \longrightarrow 1-\alpha\,,$$

  with z the $(1-\alpha/2)$-quantile of $\mathcal{N}(0,1)$ ($z = 1.96$ at 95%): the Monte Carlo estimate, of width $\propto 1/\sqrt{N}$.

- A decision obtained *from* the sample depends on it: unbiasedness does not apply.

Biased estimator

Theorem (Downward and decreasing bias)

Let $\mathcal{Y}_1$ be nonempty, let $(\boldsymbol{\xi}^n)_{n\in\mathbb{N}}$ be independent copies of $\boldsymbol{\xi}$, and assume $\widehat{v}_N$ integrable for every N. Then, for every N,

$$\mathbb{E}_{\mathbb{P}^{\otimes N}}\left[\widehat{v}_N\right] \;\leq\; \mathbb{E}_{\mathbb{P}^{\otimes(N+1)}}\left[\widehat{v}_{N+1}\right] \;\leq\; v^{\mathrm{SP}} \,.$$

Proof of the bound:

- Fix $y_1 \in \mathcal{Y}_1$: $\widehat{J}_N(y_1) \geq \widehat{v}_N$ for every draw.
- Take expectations; $\widehat{J}_N(y_1)$ is unbiased:

$$J(y_1) = \mathbb{E}_{\mathbb{P}^{\otimes N}}\left[\widehat{J}_N(y_1)\right] \geq \mathbb{E}_{\mathbb{P}^{\otimes N}}\left[\widehat{v}_N\right] \,.$$

- Take the infimum over $y_1 \in \mathcal{Y}_1$: $v^{\mathrm{SP}} \geq \mathbb{E}_{\mathbb{P}^{\otimes N}}\left[\widehat{v}_N\right]$.

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Decreasing bias

Let $\widehat{J}_N^{-n}$ be the sampled objective on the N draws $(\boldsymbol{\xi}^k)_{k\in[N+1],\, k\neq n}$. Each draw appears in N of them, so

$$\widehat{J}_{N+1}(y_1) = \frac{1}{N+1}\sum_{n=1}^{N+1} \widehat{J}_N^{-n}(y_1) \qquad \text{for every } y_1 \ .$$

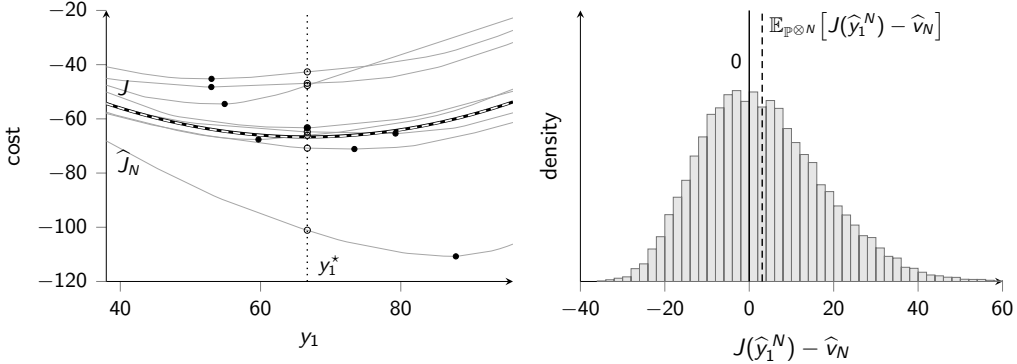
The infimum of an average is at least the average of the infima:

$$\widehat{v}_{N+1} = \inf_{y_1\in\mathcal{Y}_1} \frac{1}{N+1}\sum_{n=1}^{N+1} \widehat{J}_N^{-n}(y_1) \geq \frac{1}{N+1}\sum_{n=1}^{N+1} \inf_{y_1\in\mathcal{Y}_1} \widehat{J}_N^{-n}(y_1) \ .$$

Each $\widehat{J}_N^{-n}$ uses N independent copies of $\boldsymbol{\xi}$, so its infimum has the law of $\widehat{v}_N$; taking expectations,

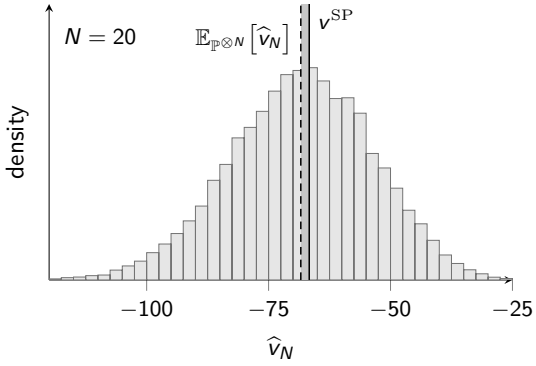
$$\mathbb{E}_{\mathbb{P}^{\otimes(N+1)}}\left[\widehat{v}_{N+1}\right] \geq \frac{1}{N+1}\sum_{n=1}^{N+1} \mathbb{E}_{\mathbb{P}^{\otimes N}}\left[\widehat{v}_N\right] = \mathbb{E}_{\mathbb{P}^{\otimes N}}\left[\widehat{v}_N\right] \ .$$

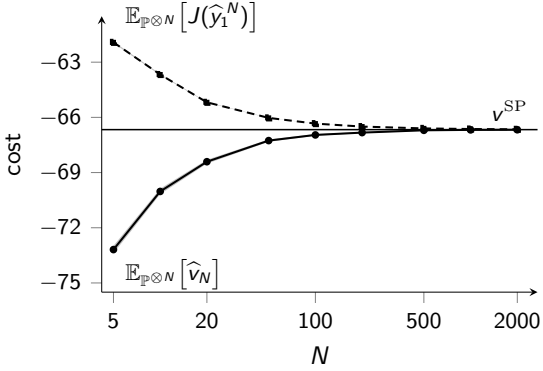
Why the minimization biases the value



Newsvendor, $c = 1$, $p = 3$, demand uniform on $[0, 100]$, $N = 20$. *Left*: eight sampled objectives $\widehat{J}_N$ (grey), each dotted at its minimum $(\widehat{y}_1^N, \widehat{v}_N)$, and their expectation J (black); each dot lies at or below its curve's value at $y_1^\star = 200/3$ (dotted). *Right*: $J(\widehat{y}_1^N) - \widehat{v}_N$ over 20 000 runs: positive on average, yet only in 55% of runs.

The sampled value is random, and optimistic on average





Same instance, $v^{\mathrm{SP}} = -200/3$, 20 000 samples. *Left*: $\widehat{v}_N$ at $N = 20$, with v^{SP} (solid), $\mathbb{E}_{\mathbb{P}^{\otimes N}}\left[\widehat{v}_N\right]$ (dashed) and the bias shaded: the spread (standard deviation 14.7) dwarfs the bias (1.7). *Right*: as N grows, $\mathbb{E}_{\mathbb{P}^{\otimes N}}\left[\widehat{v}_N\right]$ rises to v^{SP}; above it, the expected true cost $\mathbb{E}_{\mathbb{P}^{\otimes N}}\left[J(\widehat{y}_1{}^N)\right]$ of the sampled decision.

What you must remember

- Before solving, decide how a random cost is scored, what a constraint requires when it cannot hold in every outcome, and what information is available for each decision.
- Do not replace the uncertainty by a single forecast, nor choose among scenario solutions: a deterministic problem does not value flexibility.
- Score by the expectation only if the decision is repeated many times and only the total matters.
- Beware of requiring a constraint almost surely.
- Start from the timing: for each decision, say whether it is first or second stage. Write the extensive formulation and give it to a solver first.
- Once you have solved your problem, compute the VSS.
- Build the simulator first, and judge every candidate on scenarios the optimizer never saw: never report the value of a sampled problem as the value of its solution.

What you should remember

- Three approaches to uncertainty: a set, scored by its worst case; a law, scored by its expectation; a set of laws, scored by its worst expectation.
- The value at risk considers the number of bad outcomes, not how bad they are; the average value at risk weighs them, running from the expectation to the worst case.
- A chance constraint is a value-at-risk constraint, and the average-value-at-risk constraint at the same level implies it.
- A recourse decision is a function of the observation. Under finite support the extensive formulation is deterministic, its scenario blocks coupled only through the first-stage decision.
- Under relatively complete recourse every induced constraint is already implied by the first-stage constraints.
- More information relaxes the problem: $v^{\mathrm{PI}} \leq v^{\mathrm{SP}} \leq v^{\mathrm{NA}}$. When v^{SP} is finite, EVPI and VSS are nonnegative, and measure the data and the model.
- The value of a sampled problem is optimistic on average, less so as the sample grows.

Questions?



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